

DATA ENGINEERING PROJECT REPORT ON “STUDENT ATTENDANCE TRACKING SYSTEM”

Name: Abhinav Singh
Roll Number: 23051398
Batch: Data Engineering
Program: Industry Elective
Department: Computer Science and Engineering

PROBLEM STATEMENT

Traditional attendance systems are often manual, error-prone, and lack real-time tracking capabilities. They do not scale efficiently for large datasets and fail to provide meaningful analytical insights such as attendance trends, defaulter identification, and performance monitoring.

There is a need for an automated system that can capture attendance data in real time, process it efficiently, and generate actionable insights. This project addresses these challenges by developing a real-time attendance tracking system using a streaming data pipeline, ensuring accuracy, scalability, and data-driven decision-making.

SOLUTION & FEATURES

To overcome the limitations of traditional systems, this project proposes an Attendance Tracking System based on real-time data streaming and distributed processing. The system is designed as an end-to-end data pipeline that automates data ingestion, processing, analysis, and visualization.

Solution Overview

The system captures attendance data from CSV input and streams it through a pipeline consisting of Kafka, Spark, and Airflow. The processed data is then analyzed using SQL queries and visualized through dashboards.

Key Features

1. Real-Time Data Streaming
 - Attendance data is streamed using Apache Kafka
 - Enables continuous and real-time data flow
2. Automated Workflow Orchestration
 - Apache Airflow manages the execution of pipeline tasks
 - Ensures proper sequencing (Zookeeper → Kafka → Producer → Spark)

3. Scalable Data Processing

- Apache Spark Streaming processes large volumes of data efficiently
- Supports distributed computation

4. Structured Data Management

- Data is transformed into a clean and structured format
- Ensures consistency and reliability

5. Advanced SQL Analytics

- Multi-level SQL queries (Basic, Intermediate, Advanced)
- Generates insights such as:
 - Attendance percentage
 - Student ranking
 - Defaulter detection
 - Course-wise performance

6. Visualization Dashboard

- Graphical representation of insights using Python (Matplotlib)
- Includes trends, comparisons, and KPI summaries

System Workflow

The system follows a streaming pipeline architecture:

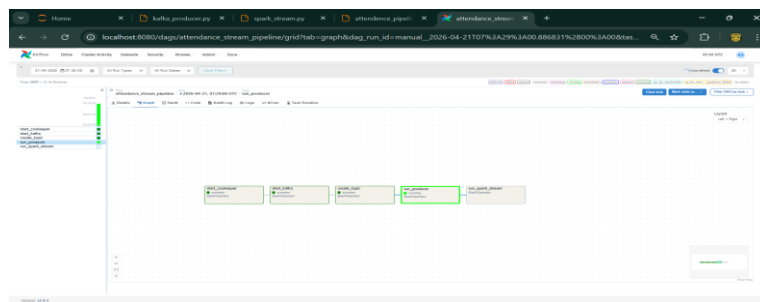
CSV Data → Kafka Producer → Kafka Topic → Spark Streaming → Processed Data →
SQL Analysis → Dashboard

↑
Airflow DAG (Automation)

This workflow ensures smooth data movement from ingestion to visualization.

SCREENSHOTS

● Airflow DAG Execution



This screenshot shows the successful execution of the Airflow DAG orchestrating the pipeline. All tasks—starting Zookeeper, Kafka, topic creation,

producer execution, and Spark streaming—are executed in sequence, indicating correct dependency management and automation.

- Kafka Producer(Streaming Data)

```
kiit@Abhinav: $ cd /mnt/c/Users/KIIT0001/OneDrive/Desktop/at
kiit@Abhinav: /mnt/c/Users/KIIT0001/OneDrive/Desktop/attend
python3 scripts/kafka_producer.py
Starting Kafka Producer...
Sent: {'student_id': 12648, 'course_id': 1960, 'attendance_d
ate': '2026-03-01', 'status': 'Present'}
Sent: {'student_id': 25218, 'course_id': 1670, 'attendance_d
ate': '2026-03-02', 'status': 'Absent'}
Sent: {'student_id': 42995, 'course_id': 1496, 'attendance_d
ate': '2026-03-04', 'status': 'Present'}
Sent: {'student_id': 46889, 'course_id': 575, 'attendance_da
te': '2026-03-10', 'status': 'Present'}
Sent: {'student_id': 12673, 'course_id': 858, 'attendance_da
te': '2026-03-08', 'status': 'Present'}
Sent: {'student_id': 24108, 'course_id': 735, 'attendance_da
te': '2026-03-08', 'status': 'Absent'}
Sent: {'student_id': 34055, 'course_id': 1974, 'attendance_d
ate': '2026-03-08', 'status': 'Absent'}
Sent: {'student_id': 36728, 'course_id': 1587, 'attendance_d
ate': '2026-03-01', 'status': 'Present'}
Sent: {'student_id': 41307, 'course_id': 1057, 'attendance_d
ate': '2026-03-01', 'status': 'Absent'}
Sent: {'student_id': 23744, 'course_id': 794, 'attendance_da
te': '2026-03-10', 'status': 'Present'}
Sent: {'student_id': 4857, 'course_id': 1202, 'attendance_da
te': '2026-03-03', 'status': 'Present'}
Sent: {'student_id': 15930, 'course_id': 618, 'attendance_da
te': '2026-03-06', 'status': 'Present'}
Sent: {'student_id': 31247, 'course_id': 1974, 'attendance_d
```

The Kafka producer streams attendance records from a CSV file into a Kafka topic. Each message represents an attendance event (student_id, course_id, date, status), demonstrating real-time data ingestion.

- Spark Streaming Output

```
r[numRows=20, truncate=false]] is committing.
Batch: 0
-----
|student_id|course_id|attendance_date|status|
-----+-----+-----+-----+
|12648|1960|2026-03-01|Present|
|25218|1670|2026-03-02|Absent|
|42995|1496|2026-03-04|Present|
|46889|575|2026-03-10|Present|
|12673|858|2026-03-08|Present|
|24108|735|2026-03-08|Present|
|34055|1974|2026-03-08|Absent|
|36728|1587|2026-03-01|Present|
|41307|1057|2026-03-01|Absent|
|23744|794|2026-03-10|Present|
|4857|1202|2026-03-03|Present|
-----+-----+-----+-----+
26/04/20 23:01:37 INFO WriteToDataSourceV2Exec: Data source
write support MicroBatchWrite[epoch: 0, writer: ConsoleWrite
```

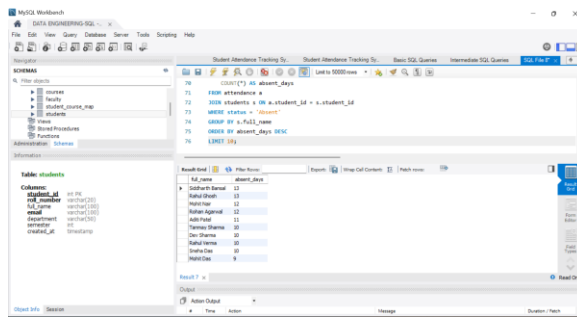
```
Batch: 1
26/04/20 23:03:21 INFO CodeGenerator: Code generated in 12.4
78807 ms
26/04/20 23:03:22 INFO CodeGenerator: Code generated in 24.4
71373 ms
-----
|student_id|course_id|attendance_date|status|
-----+-----+-----+-----+
|12648|1960|2026-03-01|Present|
|25218|1670|2026-03-02|Absent|
|42995|1496|2026-03-04|Present|
|46889|575|2026-03-10|Present|
|12673|858|2026-03-08|Present|
|24108|735|2026-03-08|Present|
|34055|1974|2026-03-08|Absent|
|36728|1587|2026-03-01|Present|
|41307|1057|2026-03-01|Absent|
|23744|794|2026-03-10|Present|
|4857|1202|2026-03-03|Present|
-----+-----+-----+-----+
```

Spark consumes the Kafka stream and processes the data into a structured format. This step performs parsing, basic cleaning, and prepares the dataset for downstream SQL analysis.

- SQL Query Output(Analytics)

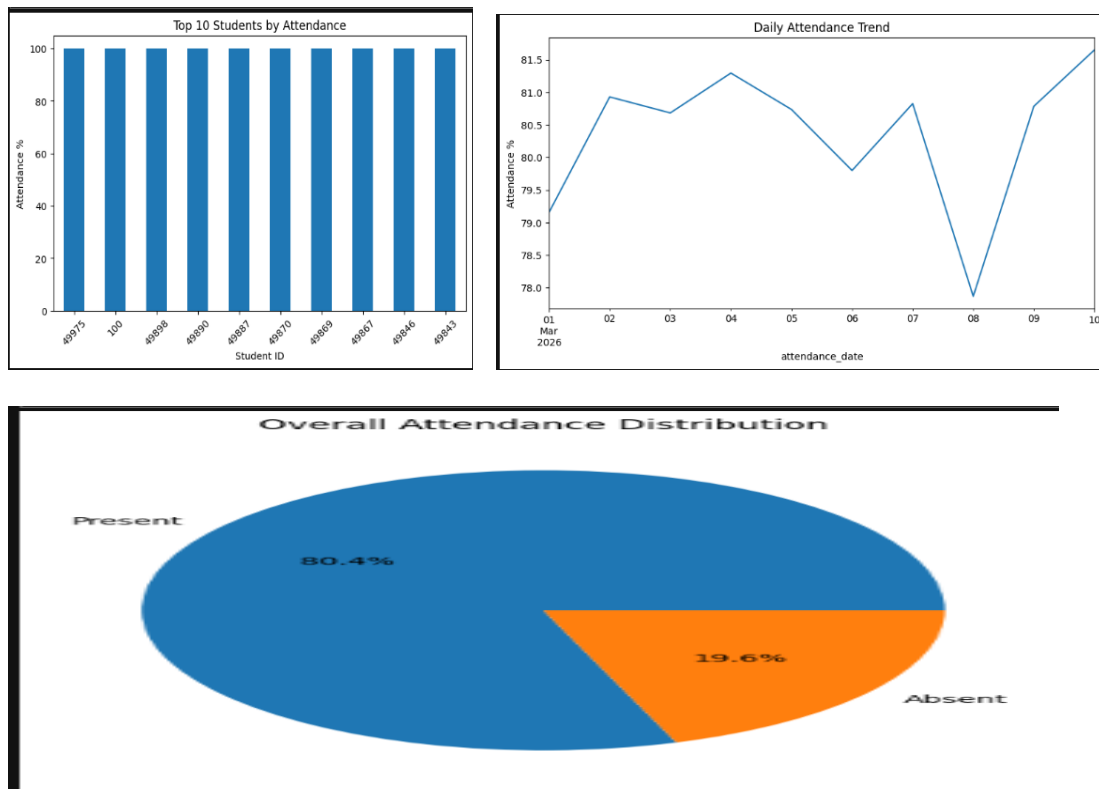
The screenshot shows a SQL query result in DBeaver. The query is: `SELECT student_id, course_id, attendance_date, status FROM attendance`. The result is displayed in a table with 4 columns: student_id, course_id, attendance_date, and status. The data is sorted by student_id.

student_id	course_id	attendance_date	status
12648	1960	2026-03-01	Present
25218	1670	2026-03-02	Absent
42995	1496	2026-03-04	Present
46889	575	2026-03-10	Present
12673	858	2026-03-08	Present
24108	735	2026-03-08	Present
34055	1974	2026-03-08	Absent
36728	1587	2026-03-01	Present
41307	1057	2026-03-01	Absent
23744	794	2026-03-10	Present
4857	1202	2026-03-03	Present



This output demonstrates analytical queries such as attendance percentage calculation, student ranking, and defaulter identification. It validates the correctness of aggregations, joins, and window functions used in the project.

● Attendance Dashboard



The dashboard visualizes key insights such as top-performing students, overall attendance percentage, and daily attendance trends, enabling clear analysis of student performance and patterns.

TECH STACK

The system is built using modern data engineering tools to enable real-time data processing, automation, and analytics.

- Python: Used for Kafka producers, Spark jobs, and visualization.
- Apache Kafka: Enables real-time data streaming.
- Apache Spark: Processes streaming data efficiently.

- Apache Airflow: Automates pipeline execution.
- SQL (MySQL): Performs analytical queries.
- Matplotlib & Jupyter: Used for visualization.
- Git & GitHub: Version control and repository management.

UNIQUE POINTS

This project stands out due to its integration of modern data engineering technologies to build a complete and scalable attendance tracking system.

- **End-to-End Data Pipeline:**
The system implements a full pipeline from data ingestion to visualization, ensuring seamless flow of data across multiple stages.
- **Real-Time Streaming Architecture:**
Unlike traditional batch systems, this project uses Apache Kafka and Spark Streaming to process attendance data in real time, enabling instant data availability.
- **Workflow Automation using Airflow:**
The entire pipeline is automated using Apache Airflow, ensuring proper task scheduling, dependency management, and reduced manual intervention.
- **Advanced SQL Analytics:**
The project includes multi-level SQL queries, including aggregation, joins, and window functions, to extract meaningful insights such as attendance percentage, ranking, and defaulter identification.
- **Insight-Driven Dashboard:**
The dashboard focuses on key analytics such as top-performing students, overall attendance, and daily trends, providing a clear understanding of attendance patterns.
- **Scalable and Modular Design:**
The architecture is designed to handle increasing data volume efficiently and can be extended for real-world deployment scenarios.

FUTURE IMPROVEMENTS

Although the current system successfully implements a real-time attendance tracking pipeline, several enhancements can be made to improve functionality, scalability, and usability.

- **Cloud Deployment:**
The system can be deployed on cloud platforms such as AWS or Azure to improve scalability, reliability, and accessibility for large-scale usage.
- **Real-Time Interactive Dashboard:**
Integration with tools like Streamlit or Power BI can enable interactive and live dashboards, allowing users to monitor attendance in real time.
- **Machine Learning Integration:**
Predictive models can be developed to analyze attendance patterns and forecast student behavior, helping in early identification of potential defaulters.
- **Web/Mobile Application Interface:**
A user-friendly interface can be developed for students and faculty to mark attendance and view analytics in real time.
- **Authentication and Role-Based Access:**
Security can be enhanced by implementing login systems and role-based access control for administrators, faculty, and students.
- **Database Optimization:**
Indexing, partitioning, and query optimization techniques can be applied to improve performance for large datasets.